



Research Paper

Sustainable methanol production from CO₂ hydrogenation: Boosting productivity and profitability using dynamic multi-stage cooling strategyMaedeh Afzali, Mohammad Hasan Khademi^{*}, Mahboubeh Toghyani

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ABSTRACT

CO₂ emissions pose a critical environmental threat, contributing to global warming and climate change. The efficient hydrogenation of CO₂ to methanol is a critical strategy for captured carbon, providing a sustainable pathway to reduce greenhouse gases. This study evaluates two comprehensive approaches, a simple quasi-steady-state (QSS) model and a complex dynamic model, to boost methanol production efficiency from CO₂ hydrogenation in a multi-tubular fixed-bed reactor by identifying optimal cooling strategies in single- and multi-stage systems. Key operating parameters, including pressure and H₂/CO₂ ratio, are distinguished as pivotal to shifting equilibrium toward higher methanol yields, while a tailored cooling system can effectively manage CO₂ hydrogenation reactions. At first, the optimal temperature profile along the reactor's length is determined using both models. The comparison of the results showed that both models exhibit similar behavior and are in good agreement with industrial data. Subsequently, a time-stepwise strategy is employed to achieve an optimal one-stage cooling system over 4 years. Results indicated that the QSS model offers a practical and cost-effective approach compared to the dynamic model, achieving consistent temperature control, particularly in multi-stage cooling systems. Meanwhile, the dynamic model adjusts coolant temperatures across the catalyst's lifespan, achieving up to a 9.95 % increase in methanol yield under declining catalyst activity. Economic analysis reveals substantial revenue enhancements due to applying the optimal two-stage cooling strategy—up to \$144.3 million over 4 years, underscoring the two-stage cooling system with higher performance as the promising potential to increase profitability in industrial CO₂ hydrogenation. In conclusion, this study provides valuable insights into effective temperature management strategies to maximize methanol production in the presence of catalyst deactivation.

1. Introduction

Currently, emissions of greenhouse gases, including carbon dioxide (CO₂), are among the environmental concerns that received considerable attention. Therefore, numerous studies have concentrated on advanced and innovative technologies aimed at capturing, utilizing, and storing carbon [1–3]. It has been suggested that the conversion of CO₂ into valuable chemicals via various processes, to reduce greenhouse gas emissions, could serve as a beneficial solution. For example, photocatalytic, electrocatalytic, and photoelectrocatalytic reduction of CO₂ are considered promising processes for the future [4,5]. However, current techniques are still unable to fulfill the demands of large-scale industrial applications. Currently, the thermo-catalytic approach remains the most practical option for industrial use.

Within the thermo-catalytic approach, the hydrogenation of CO₂ to

methanol has become a significant attraction for the mitigation of greenhouse gas emissions [6]. Globally, the demand for CO₂ in the synthesis of methanol amounts to 8 million tonnes annually [7]. It is important to emphasize that currently, about 90 % of methanol produced worldwide is derived from natural gas through a two-step process (natural gas reforming to syngas and syngas to methanol) [8], not yet through the direct hydrogenation of CO₂ [9]. Nevertheless, the direct hydrogenation of CO₂ to methanol is less exothermic compared to the conversion of synthesis gas to methanol, resulting in fewer by-products and lower emissions [10]. In addition, producing methanol from CO₂, as a promising technology, offers advantages from both economic (significant reduction in the natural gas consumption and process equipment) and environmental (reducing CO₂ emissions) perspectives.

On the other hand, the CO₂ hydrogenation to methanol can be classified as a “green” or environmentally friendly process, particularly when the hydrogen used is sourced from sustainable and renewable

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Nomenclature			
a	Catalyst activity	P	Bed pressure, <i>bar</i>
$\bar{a}$	Relative activity of the catalyst	$\dot{Q}_{cw}$	Volumetric flow rate of cooling water, $m^3 s^{-1}$
$\langle \bar{a} \rangle$	Average relative activity of the catalyst	r_j	Rate of reaction j , $mol\ kg_{cat}^{-1} s^{-1}$
A	Cross section area, m^2	R	Universal gas constant, $J\ mol^{-1} K^{-1}$
$C_{p,g}$	Heat capacity of the gas mixture, $J\ kg^{-1} K^{-1}$	Re	Reynold number
$C_{p,cat}$	Heat capacity of catalyst, $J\ kg^{-1} K^{-1}$	t	Time, <i>s</i>
$C_{p,cw}$	Heat capacity of cooling water, $J\ kg^{-1} K^{-1}$	T	Bed temperature, <i>K</i>
d_p	Catalyst diameter, <i>m</i>	$T_{cooling}$	Cooling temperature, <i>K</i>
d_t	Tube diameter, <i>m</i>	T_{ref}	The reference temperature, <i>K</i>
E_{cw}	Thermal energy for warming the cooling water, $J\ s^{-1}$	u	Superficial gas velocity, $m\ s^{-1}$
E_{el}	Electrical energy, <i>kWh</i>	U	Overall heat transfer coefficient, $W\ m^{-2} K^{-1}$
E_{ph}	Thermal energy for preheating the feed gas, $J\ s^{-1}$	z	Axial coordinate, <i>m</i>
f_i	Fugacity of component <i>i</i> , <i>bar</i>	Greek letters	
k_j	The rate constant of reaction j , $mol\ kg^{-1} s^{-1} bar^{-1/2}$	ΔH_j^r	The heat of reaction j , $J\ mol^{-1}$
K_i	Adsorption equilibrium constant of component <i>i</i> , bar^{-1}	ΔP	Pressure difference, <i>Pa</i>
K_{pj}	Equilibrium constant of reaction j	ν_{ij}	Stoichiometric coefficient of component <i>i</i> in reaction j
L	Length of each stage, <i>m</i>	ρ_{cat}	Catalyst density, $kg\ m^{-3}$
$\dot{m}$	Total mass flow rate, $kg\ s^{-1}$	ω_i	Mass fraction of component <i>i</i>
$\dot{m}_{cw}$	Mass flow rate of cooling water, $kg\ s^{-1}$	ε	Bed porosity
MW_i	The molecular weight of component <i>i</i>	ρ_g	Density of gas mixture, $kg\ m^{-3}$
N	Number of tubes	λ_f	Latent heat of water, $J\ kg^{-1}$

methods such as chlor-alkali plants [11], and solar/wind-based water electrolysis [12–14] using an alkaline or a proton exchange membrane electrolyzer. Similarly, CO₂ can be captured from various sources including the atmosphere, steel mills, paper mills, cement industry, petrochemicals, or power plants [12,15].

From the technological standpoint, the CO₂ hydrogenation to methanol is stated at technology readiness levels (TRLs) 6–7 [13], which is based on the syngas to methanol process as a mature technology. In this regard, diverse research on kinetic model development [16,17], process intensification [18,19], mathematical modeling [10,20–22], process simulation and optimization [23–25], catalyst modification [26–29], and techno-economic and environmental assessment [30–32] have been carried out to promote this TRL.

Among various Cu/noble metal/metal oxide/composite-based catalysts, Cu/ZnO/Al₂O₃ is the most dominant catalyst for CO₂ hydrogenation to methanol due to its long-lasting operational durability (life span of 2–5 years [33]), remarkable resistance to poisoning, affordability, and outstanding catalytic activity [34]. However, the primary concerns for researchers aiming to develop this process industrially are the oxidation and deactivation of the catalyst, which occur due to the presence of CO₂ and the production of water during the reaction, respectively [33]. Additionally, this process suffers from low methanol selectivity and CO₂ conversion in once-through reactors due to equilibrium constraints at high temperatures [29].

To address these challenges, researchers have focused on developing reactor configurations or adding external equipment to the process to boost the CO₂ hydrogenation. Given that CO₂ is thermodynamically more stable than CO, two strategies have been proposed to boost the CO₂ conversion. The first strategy involved recycling the unconverted gases including CO, H₂, and CO₂ to the entrance of the reactor [10,21,35]. The second strategy was a two-step process known as the CAMERE process, which consisted of the reverse water-gas shift (RWGS) reactor, where CO₂ was transformed into CO, followed by a methanol synthesis reactor, in which syngas was converted into methanol [36–38]. Furthermore, researchers suggested various technologies to overcome thermodynamic equilibrium constraints and delay in catalyst deactivation, such as water removal from the conventional CO₂ hydrogenation reactor or the

CAMERE process by employing a flash drum after each reactor [10,39], equipping the reactor by a H₂O-permselective membrane [37,39,40], designing a sorption-enhanced reactor [41], or developing a bifunctional catalytic membrane reactor [42]. These concepts resulted in higher methanol selectivity and CO₂ conversion.

Given that the CO₂ hydrogenation process generates heat due to its exothermic nature, another main reactor design issue is to effectively dissipate the heat produced during the hydrogenation, ensuring that the reactor maintains in isothermal conditions. The heat generated inside the tubes of the conventional methanol synthesis reactor is extracted by boiling water, which flows as a coolant in the reactor jacket. Samimi et al. [43] investigated the phase stability during the hydrogenation of CO₂ in various tubular reactors, including those cooled by gas, water, and a combination of gas and water. Their findings indicate that the reactor design employing gas coolant is more effective in enhancing methanol yield and CO₂ conversion. However, evaluating the three types of reactors including gas-cooled, water-cooled, and adiabatic reactors for the CO₂ hydrogenation process by Cui and Kær [44] showed that the water-cooled reactor includes benefits such as a broader range of inlet temperatures for better control, low hot-spot temperature, extended catalyst lifespan, effective heat removal, and energy recovery in the coolant. Instead, the gas-cooled and adiabatic reactors showed relatively medium and low levels of performance and capital cost, respectively. This suggests that these two reactor types could be suitable for small-scale CO₂ hydrogenation to methanol processes.

Leonzio et al. [10] conducted a sensitivity analysis on CO₂ hydrogenation, revealing that the temperature of the cooling medium plays a more critical role in enhancing methanol selectivity and CO₂ conversion than does the overall heat transfer coefficient. Additionally, various researchers have optimized the reactor's inlet temperature, alongside other parameters, to maximize methanol yield and CO₂ conversion [20,45], minimize the methanol production cost [46], maximize the process profit [47], and minimize the energy consumption [48].

Due to the significant impact of temperature on the reaction kinetics, thermodynamic equilibrium, and catalyst deactivation, it is crucial to establish an optimal temperature policy for the efficient operation of a CO₂ hydrogenation reactor. To the best of the authors' knowledge, no

detailed and comprehensive study has been found on the evaluation of optimal cooling system temperature of this reactor in the presence of catalyst deactivation. During a 4-year operational period, the catalyst's activity diminishes to a level where it is no longer economically viable to continue running the reactor; this gradually reduces methanol production. In this regard, the concept of developing the CO₂ hydrogenation reactor with a dynamic multi-stage cooling system is employed to boost the productivity and profitability of methanol over a 4-year operational period. To achieve this goal, two distinct strategies are employed.

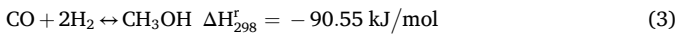
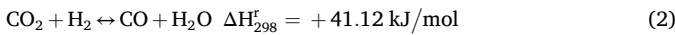
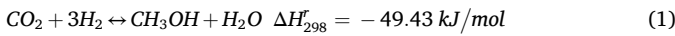
The first strategy, as a theoretical solution, aims to determine the optimal temperature profile along the length of the reactor's cooling jacket through two distinct models—the QSS model and the dynamic model—to maximize the rate of methanol production. In fixed-bed methanol synthesis reactors, temperature and concentration profiles reach steady conditions within a few seconds because of high heat-transfer rates and small tube diameters, while catalyst deactivation occurs over much longer periods—from days to months. In other words, the internal time constants—the residence time, the reaction time constant, and the thermal time constant—are much smaller than the deactivation time constant. It seems that this clear separation of time-scales makes it possible to neglect the dynamic terms in the model when determining optimal operating policies. However, by comparing these two models, it can be confirmed whether the simple QSS model could serve as an effective substitute for the more complex dynamic model and significantly reduce the computational effort.

Following the findings of the first strategy, the second strategy deals with the design of a dynamic multi-stage cooling system that uses various constant cooling temperatures over distinct time periods as a practical solution. A heuristic optimization is conducted for both strategies at specific levels of catalyst activity. The varying levels of catalyst activity are chosen by an unsteady-state model to represent the reactor dynamics over the lifetime of the catalyst. Finally, an economic analysis is highlighted to select the optimal number of cooling stages over a 4-year period.

2. Methodology

2.1. Kinetics model

Methanol is produced from carbon dioxide and hydrogen at pressures ranging from 5 to 10 MPa and temperatures between 493 and 603 K over industrial Cu/ZnO/Al₂O₃ catalysts [35]. The primary reactions involved in the synthesis of methanol include CO₂ hydrogenation, reverse water-gas shift, and CO hydrogenation as follows:



The kinetics rate expressions corresponding to the reactions (1)–(3), given by Graaf et al. [49], are applied as follows:

$$r_1 = \frac{k_1 K_{\text{CO}_2} [f_{\text{CO}_2} f_{\text{H}_2}^{1.5} - f_{\text{CH}_3\text{OH}} f_{\text{H}_2\text{O}} / (f_{\text{H}_2}^{1.5} K_{p1})]}{(1 + K_{\text{CO}} f_{\text{CO}} + K_{\text{CO}_2} f_{\text{CO}_2}) [f_{\text{H}_2}^{0.5} + (K_{\text{H}_2\text{O}} / K_{\text{H}_2}^{0.5}) f_{\text{H}_2\text{O}}]} \quad (4)$$

$$r_2 = \frac{k_2 K_{\text{CO}_2} [f_{\text{CO}_2} f_{\text{H}_2} - f_{\text{CO}} f_{\text{H}_2\text{O}} / K_{p2}]}{(1 + K_{\text{CO}} f_{\text{CO}} + K_{\text{CO}_2} f_{\text{CO}_2}) [f_{\text{H}_2}^{0.5} + (K_{\text{H}_2\text{O}} / K_{\text{H}_2}^{0.5}) f_{\text{H}_2\text{O}}]} \quad (5)$$

$$r_3 = \frac{k_3 K_{\text{CO}} [f_{\text{CO}} f_{\text{H}_2}^{1.5} - f_{\text{CH}_3\text{OH}} / (f_{\text{H}_2}^{0.5} K_{p3})]}{(1 + K_{\text{CO}} f_{\text{CO}} + K_{\text{CO}_2} f_{\text{CO}_2}) [f_{\text{H}_2}^{0.5} + (K_{\text{H}_2\text{O}} / K_{\text{H}_2}^{0.5}) f_{\text{H}_2\text{O}}]} \quad (6)$$

where the temperature dependence of K_{pj} , K_i , and k_j , which are referred

to as the reaction equilibrium constant, adsorption equilibrium constant, and reaction rate constant, respectively, were presented by Graaf et al. [49,50].

2.2. Catalyst deactivation model

The deactivation model of the Cu/ZnO/Al₂O₃ catalyst based on thermal sintering is taken from Hanken [51]:

$$\frac{da}{dt} = -0.00439 (h^{-1}) \exp \left(\frac{-91270 (J \text{ mol}^{-1})}{R} \left(\frac{1}{T} - \frac{1}{513 (K)} \right) \right) a^5 \quad (7)$$

at $t = 0$ $a = 0.4$
 $\tilde{a} = 1 - \frac{0.4 - a}{0.4}$

An initial activity of 0.4 was considered due to the rapid deactivation observed initially. However, a relative activity, $\tilde{a}$, scaled to start at 1.0, was applied to determine the actual reaction rate. In contrast to other deactivation models studied by various researchers, this model has been fitted with industrial data, making it the sole candidate for modeling such industrial reactors.

2.3. Mathematical model

A one-dimensional dynamic pseudo-homogeneous model is developed to estimate the catalyst activity profile over time. Furthermore, a steady-state model is derived by removing all time-derivative terms from the partial differential equations (PDEs) to specify the temperature and concentration profiles inside the catalytic bed. The heat and mass balance equations are written across a differential element created by incrementing Δz along the z unit vector. A schematic diagram of a differential element is demonstrated in Fig. 1. In this regard, the following assumptions are provided:

- A pseudo-homogeneous model is considered because the gradients of concentration and temperature between the gas and catalysts can be ignored due to higher mass and heat transfer rates compared to the reaction rate. This assumption is also supported by other studies [39,52,53].
- The compressibility factor was found to be 1.02 using the Soave–Redlich–Kwong equation of state, indicating minor deviations from ideality. Therefore, the reaction mixture is assumed to follow the ideal gas law.
- The effectiveness coefficient is set equal to 1.0 due to the lack of difference between the comparative scenarios with and without the inclusion of effectiveness factors found by Leonzio [54].
- The heat resistance of the solid wall and cooling fluid is far less significant than gas phase convective heat transfer resistance, due to the wall's high thermal conductivity and the high convective heat transfer coefficient of circulating boiling water [54], allowing them to be neglected.
- The phenomenon of axial dispersion is not taken into account due to the mass and thermal Peclet numbers being large enough

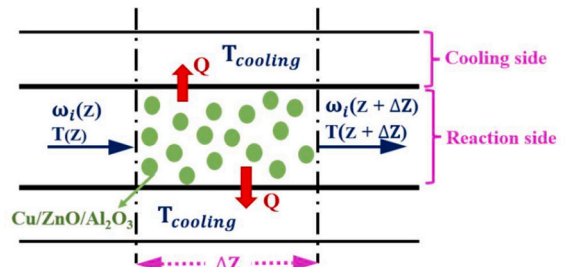


Fig. 1. Schematic diagram of the differential element.

($Pe_m = 5371$ and $Pe_t = 9854$, both more than the criterion value of 100 [55]).

- Leonzio and Foscolo [21] observed a reasonable consistency between one- and two-dimensional models for a multi-tubular reactor with tubes of 50.8 mm diameter. Accordingly, the one-dimensional model is valid for tubes with smaller diameters and is employed in this study ($d_t = 38$ mm).

Partial mass balance for component i

$$\rho_g \varepsilon \frac{\partial \omega_i}{\partial t} = -\frac{\dot{m}}{A} \frac{\partial \omega_i}{\partial z} + \tilde{a} \rho_{cat} (1 - \varepsilon) MW_i \sum_{j=1}^3 v_{ij} r_j$$

at $z = 0$ $\omega_i = \omega_{i,0}$

at $t = 0$ $\omega_i = 0$

Energy balance

$$\left\{ \rho_g \varepsilon C_{p,g} + \rho_{cat} (1 - \varepsilon) C_{p,cat} \right\} \frac{\partial T}{\partial t} = -\frac{\dot{m} C_{p,g}}{A} \frac{\partial T}{\partial z} + \tilde{a} \rho_{cat} (1 - \varepsilon) \sum_{j=1}^3 r_j (-\Delta H_j^f) + \frac{\pi d_t}{A} U (T_{cooling} - T)$$

at $z = 0$ $T = T_0$

at $t = 0$ $T = T_0$

Pressure drop

The pressure drop along the catalytic bed is described by the Ergun equation.

$$\frac{dP}{dz} = - \left(1.75 + 150 \left(\frac{1 - \varepsilon}{Re} \right) \right) \frac{\rho u^2}{d_p} \left(\frac{1 - \varepsilon}{\varepsilon^3} \right)$$

at $z = 0$ $P = P_0$

2.4. Economic evaluation

A thorough economic evaluation of the CO₂ hydrogenation process requires detailed data from the process and reactor configurations, typically obtained during the detailed design phase. By calculating and comparing critical economic parameters, it is possible to gain a better understanding of the operational costs and potential revenues associated with different designs, allowing for rough estimates of their performance. This evaluation underscores the importance of utility costs and gross revenue calculations in determining overall profitability.

To calculate the utility cost, Eq. (11) provides the flow rate of cooling water, $\dot{m}_{cw}$, for removing the amount of heat released during CO₂ hydrogenation reactions along the reactor's length:

$$\dot{m}_{cw} = \frac{1}{\lambda_f} \int_0^L N \pi d_t U (T - T_{cooling}) dz$$

where λ_f refers to the latent heat of water at the shell temperature.

The pump's electrical energy, E_{el} , to boost the cooling water pressure to a certain level for the reactor shell is determined using Eq. (12):

$$E_{el} = \frac{\dot{Q}_{cw} \Delta P}{\eta \times 1000} \times \text{Operating time}$$

where η is the pump efficiency, assumed to be 0.7.

The thermal energy, E_{cw} , required to warm the cooling water from the ambient temperature (i.e., 298 K) to the shell temperature using natural gas as fuel is calculated as follows:

$$E_{cw} = \int_{298}^{T_{cooling}} \dot{m}_{cw} C_{p,cw} dT$$

The thermal energy, E_{ph} , required to preheat the gas feed from the reference temperature (i.e., 503 K) to the optimal one is determined as follows:

$$E_{ph} = \int_{T_{ref}}^{T_0^{opt}} \dot{m} C_{p,g} dT$$

The electrical and thermal energy costs are then calculated through Eqs. (15) and (16), respectively.

$$\text{Electrical Energy Cost} = \text{Electricity price} \times E_{el}$$

$$\text{Thermal Energy Cost} = \text{Natural gas price} \times (E_{cw} + E_{ph}) \times \text{Operating time}$$

In the literature, the costs of natural gas, electricity, and water are

reported as \$3.16 per GJ [56], \$0.063 per kWh [57], and \$0.62 per tonne [57], respectively. These prices are updated based on the Chemical Engineering Plant Cost Index (CEPCI) for 2024 and converted to \$4.16 per GJ for natural gas, \$0.071 per kWh for electricity, and \$0.695 per tonne for water. The total utility cost is ultimately calculated by the summation of the electrical and energy costs, along with the cost of water consumption, considering an annual operating time of 8150 h.

Another important economic parameter is gross revenue, representing the total income generated from the sale of methanol produced in the process. It can be calculated using formula (17):

$$\text{Gross Revenue} = \text{Methanol Production rate} \times \text{Price of Methanol} \times \text{Operating time}$$

In this analysis, the price of methanol is estimated at \$438.72 per tonne, as reported in reference [57]. Understanding gross revenue is essential for assessing the financial viability of different reactor configurations, as it directly impacts overall profitability and helps justify the investment in specific process designs.

2.5. Numerical solution

The governing equations for the steady-state model were composed of a set of ordinary differential equations (ODEs) and algebraic equations. The ODEs were transformed into nonlinear algebraic equations using the backward finite difference formula. For this purpose, the reactor's length was segmented into 200 uniformly distributed points. All algebraic equations were solved using the "fsolve" function within the MATLAB programming environment (R2020a), which employs the trust-region-dogleg procedure. As the grid size decreases, the results show a diminishing reliance on the grid's size, suggesting that the numerical solution is converging toward the actual solution.

A set of ODEs, PDEs, and algebraic equations form the governing equations for the dynamic model. The method of line (MOL) was applied to solve these equations. After discretizing the PDEs along the axial direction (200 nodes), they were transformed into a set of ODEs that have time as the sole independent variable. A numerical solver specifically

designed for stiff ODEs, known as ode15s, was employed to solve these equations in the time direction. This solver utilizes a variable order method called the numerical differentiation formula (NDF). The relative and absolute error tolerances were set to 1E-5 and 1E-7, respectively, using the “odeset” function.

2.6. Optimization

The objective function is to maximize the rate of methanol production as much as possible by determining the optimal coolant temperature profile. The most precise optimal temperature profile is attained by optimizing the feed temperature and using 200 parameters representing coolant temperatures at 200 nodes as decision variables. The steady-state and dynamic model's equations serve as the equality constraints. There is a single inequality constraint that forces the catalyst bed temperatures to be above 493 K to ensure the reaction takes place [35,58], and below 543 K throughout the reactor to prevent significant catalyst deactivation [59] (catalyst sintering temperature was reported 573 K [37]). This constraint is enforced using the penalty function. Finally, the optimization problem is structured as follows:

$$\text{minimize } f = -\text{methanol production rate} + 10^7 \sum_{i=1}^2 G_i^2 \quad (18)$$

where

$$G_1 = \max\{0, (T - 543)\} \quad (19)$$

$$G_2 = \max\{0, (493 - T)\} \quad (20)$$

The Differential Evolution algorithm offers significant benefits over

other optimization methods, such as its ability to reach a global minimum regardless of the initial population, its reliance on a minimal number of control parameters, and its rapid convergence rate [60]. In this work, this algorithm has been employed as a promising optimization method to determine the optimal coolant temperature. The parameters for the DE algorithm are set as follows: the crossover probability constant is 1.0, the scaling factor is 0.8, and the population size is 50. The algorithm's strategy is based on employing binomial recombination, utilizing a mutation technique that involves one difference vector, and perturbing the best vector from the previous generation. The algorithm is terminated when the maximum number of iterations, which is set at 150, is reached, serving as the convergence criterion (see Appendix).

Fig. 2 illustrates the steps involved in the optimization procedure to determine the optimal coolant temperature.

3. Results and discussion

3.1. Grid independence study

Since the spatial derivatives in the steady-state and dynamic models' equations are discretized along the reactor length, a grid independence study is conducted for both model to ensure that the numerical results are not influenced by discretization choice. For the steady-state model, as depicted in Fig. 3(a), increasing the number of nodes beyond 200 results in negligible changes in the methanol production rate. Therefore, 200 nodes are selected for the reactor length.

For the dynamic model, Fig. 3(b) presents the sensitivity of the results to grid resolution at the first and last days of the operating period. As illustrated, the methanol production rate remains unchanged beyond 600 nodes. However, the relative error obtained using 200 nodes compared to that with 600 nodes is less than 0.1 %, making it the optimal choice for the reactor axial grid.

3.2. Reference case

The geometry and operating conditions of the CO₂ hydrogenation reactor are assumed to be identical to those of the syngas-based methanol synthesis reactor at the Shiraz Petrochemical Complex, which is detailed in Table 1 as the reference case. This reactor operates with a fresh catalyst that gradually undergoes deactivation over time due to various operational and chemical factors. The feed stream, comprising hydrogen and carbon dioxide with a molar ratio of 3:1, enters the multi-tubular fixed-bed reactor at an inlet temperature of 503 K, initiating the highly exothermic CO₂ hydrogenation reaction. The heat generated during this process is continuously removed via an integrated cooling system, which maintains a temperature of approximately 524 K to ensure that thermal runaway or hot-spot formation is effectively mitigated, thereby preserving the reactor's integrity and preventing catalyst degradation.

3.3. Model verification and validation

The verification and validation of the proposed model are crucial for establishing its predictive accuracy and applicability to industrial-scale operations. There is a lack of publicly available experimental data on pilot and industrial-scale reactors for methanol synthesis via CO₂ hydrogenation, especially under dynamic conditions and in the presence of catalyst deactivation. As a result, model validation can only be performed using actual plant data for syngas-based methanol production at the Shiraz Petrochemical Complex.

Although the validation in this study was performed using industrial data from syngas-based methanol synthesis, this approach is technically justified because both CO- and CO₂-based routes share the same catalytic reaction network and operate under similar thermodynamic and transport regimes. The main active components, reactor configuration, and thermal management strategy are identical, ensuring that the model

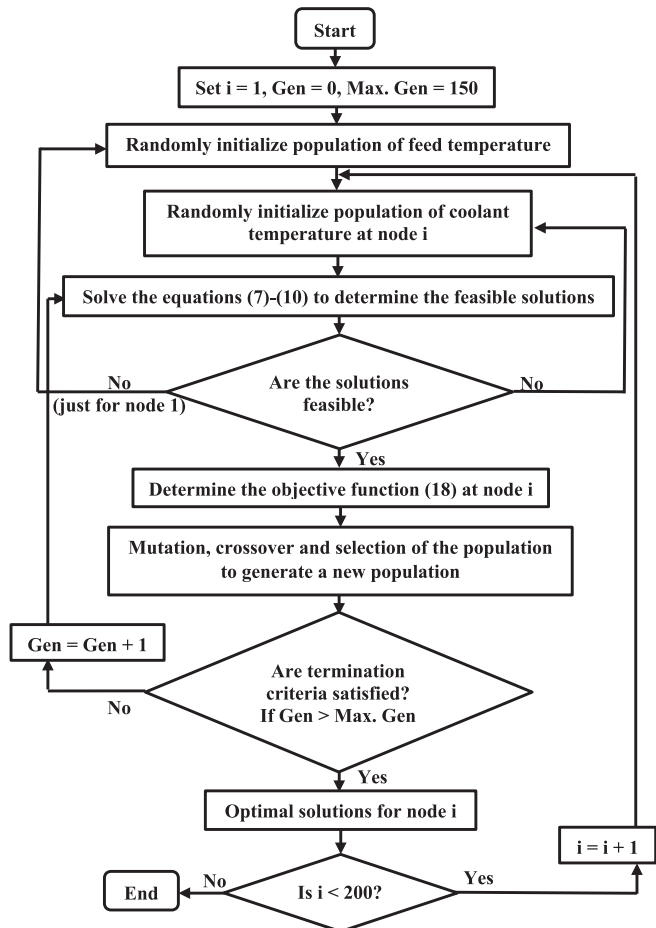


Fig. 2. Flowchart of optimization procedure.

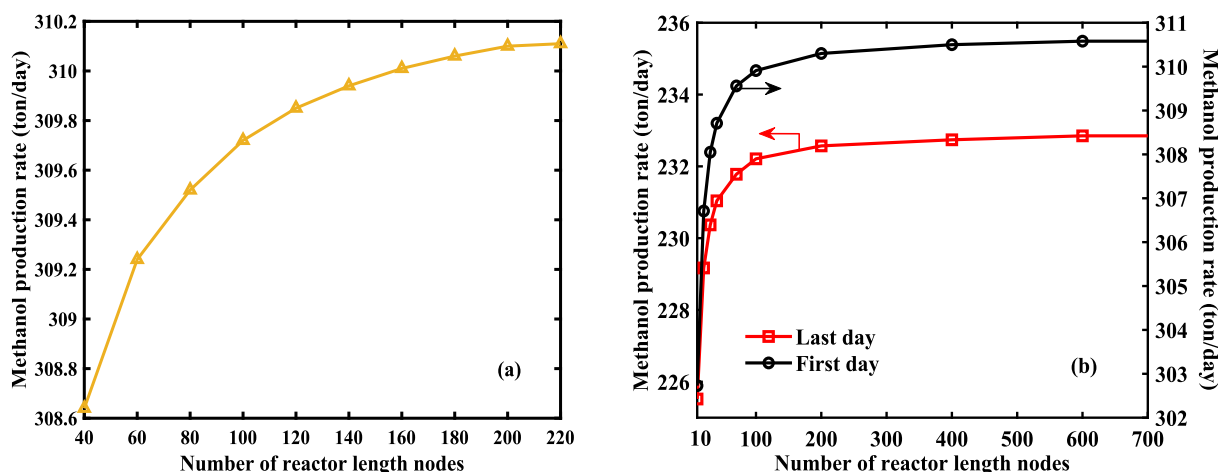


Fig. 3. Variations of methanol production rate with the number of reactor length nodes in (a) steady-state mode, and (b) dynamic mode.

Table 1

Geometry and operating conditions of the CO₂ hydrogenation reactor based on Shiraz Petrochemical Complex data [39] (Reference case).

Parameter	Value
Geometry	
Particle diameter (mm)	5.47
Particle heat capacity (kJ kg ⁻¹ K ⁻¹)	5.0
Density of catalyst bed (kg m ⁻³)	1140
Bed porosity	0.39
Reactor length (m)	7.022
Tube diameter (mm)	38
Number of tubes	2962
Operating conditions	
Inlet pressure (bar)	76.98
Feed temperature (K)	503
Cooling temperature (K)	524
Feed molar flow rate of each tube (mol s ⁻¹)	0.64
H ₂ mole fraction	0.75
CO ₂ mole fraction	0.25

Table 2

Validation of steady-state simulation results against plant data [61] for syngas-based methanol synthesis.

	Reactor inlet	Reactor outlet		Relative error (%)
		Plant data	Simulation data	
Molar flow rate (mol s ⁻¹)	0.565	0.51	0.511	0.19
Bed temperature (K)	503	528	524.06	0.74
Pressure (bar)	61.83	61.16	61.59	0.7
Composition (mol %)				
CO ₂	3.45	2.18	2.43	11.46
CO	4.66	1.44	1.51	4.86
H ₂	79.55	75.71	76.53	1.08
CH ₄	11.72	12.98	12.95	0.23
N ₂	0.032	0.16	0.035	78.12
H ₂ O	0.08	1.74	1.47	15.51
CH ₃ OH	0.032	5.49	5.05	8.01

accurately represents large-scale methanol synthesis behavior. Nevertheless, some differences must be acknowledged. CO₂ hydrogenation generally produces higher water content, involves a greater extent of the reverse water–gas shift reaction, and may induce slightly different heat release profiles and catalyst deactivation patterns compared to CO-rich feeds. Consequently, minor deviations could arise when extrapolating

Table 3

Comparison of the outlet molar flow rates of components obtained from the developed model at steady-state conditions with relevant literature data [10,62].

Species	Feed molar flow rate (mol s ⁻¹)	Outlet molar flow rate (mol s ⁻¹)		
		Model prediction	Fu et al. [62]	Leonzio et al. [10]
CO ₂	0.58	0.475	0.470	0.470
H ₂	1.88	1.586	1.600	1.590
CH ₃ OH	0	0.097	0.090	0.090
H ₂ O	0	0.107	0.110	0.110
CO	0.07	0.080	0.090	0.080

syngas-based validation results to CO₂-based operation. Future work should focus on validating the model with pilot- or demonstration-scale CO₂ hydrogenation data as such datasets become available, or through dedicated dynamic experiments designed to capture the transient thermal and kinetic effects specific to CO₂-rich feeds.

In this regard, key performance metrics, including total molar flow rate, methanol production rate, mole fractions of key components, and operational parameters such as pressure and temperature, were evaluated. Table 2 provides a comprehensive comparison between the model's steady-state predictions and the corresponding industrial plant data.

A detailed analysis of the relative error percentages, presented in Table 2, indicates that the model predictions align well with the actual plant data, demonstrating that the model is capable of accurately capturing the steady-state behavior of the methanol synthesis process. The relative errors in key parameters, such as molar flow rate, system pressure, and catalyst bed temperature, are within acceptable limits, thereby confirming the validity of the model for large-scale applications. Nevertheless, some discrepancies, particularly in the mole fractions of minor components such as N₂ and H₂O, were observed. These deviations could be attributed to several factors, including measurement uncertainties in the industrial plant or the exclusion of operational fluctuations that were not accounted for in the model. Such minor discrepancies are common in complex industrial processes where real-world conditions often differ from idealized assumptions made during model development.

Furthermore, to enhance the credibility of the model, the outlet molar flow rates of components are compared with previous studies on methanol production via CO₂ hydrogenation, showing good agreement (Table 3).

In addition to steady-state analysis, the model was also subjected to

Table 4

Validation of dynamic simulation results against plant data [52] for syngas-based methanol synthesis.

Time (day)	Methanol production rate (ton/day)		Relative error (%)
	Plant data	Simulation data	
0	295.0	316.09	7.15
100	296.5	302.04	1.87
200	302.6	293.35	3.05
300	284.3	286.74	0.86
400	277.9	281.43	1.27
500	278.2	276.87	0.47
600	253.0	272.95	7.88
700	274.0	269.53	1.63
800	286.1	266.36	6.89
900	275.5	263.53	4.34
1000	274.6	260.95	4.97
1100	262.9	258.54	1.65
1200	255.2	256.30	0.43

dynamic validation to assess its performance over time in the presence of catalyst deactivation. Table 4 presents a comparison between the syngas-based industrial methanol production on a per-day basis and the model's dynamic predictions. The dynamic mode validation is particularly important for assessing the model's capability to simulate transient phenomena, such as fluctuations in operational changes in reactor conditions. The maximum observed relative error during dynamic validation was approximately 7.88 %, which is within a reasonable margin of error for industrial process simulations. This error range further underscores the reliability of the model in predicting both steady-state and dynamic performance under varying operational scenarios. However, it is crucial to confirm the model's predictions for CO₂ hydrogenation using dynamic data from pilot plant to ensure greater accuracy.

3.4. Simulation results

Fig. 4(a) illustrates the variations of the catalyst bed temperature, methanol mass flow rate, and the cooling system temperature at the shell side of the reactor along the dimensionless length of the reactor in steady-state mode. As depicted in this figure, within the initial section of the reactor, the catalyst bed temperature rises due to the high reaction rate of the exothermic CO₂ hydrogenation process. After reaching the peak, the temperature begins to decrease as heat transferred to the shell side is greater than heat generated in the catalytic bed reactor. As the reaction progresses toward the end of the catalyst bed, the reaction rate slows down, as indicated by the reduction in the slope of the methanol mass flow rate. This reduction is linked to the depletion of reactants and

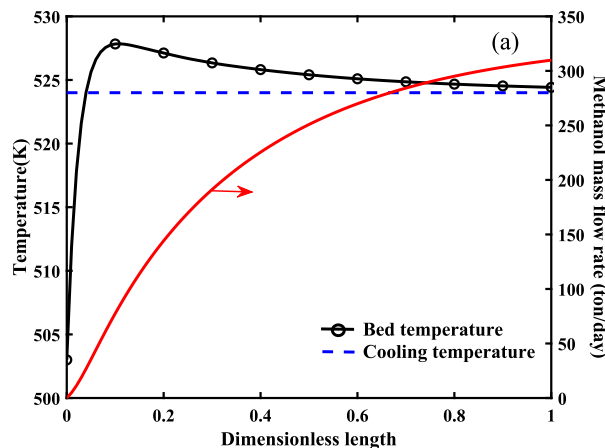


Fig. 4. (a) Catalyst bed and shell-side temperature profiles, and methanol flow rate in steady-state mode, and (b) average catalyst activity and methanol production rate profiles in dynamic mode.

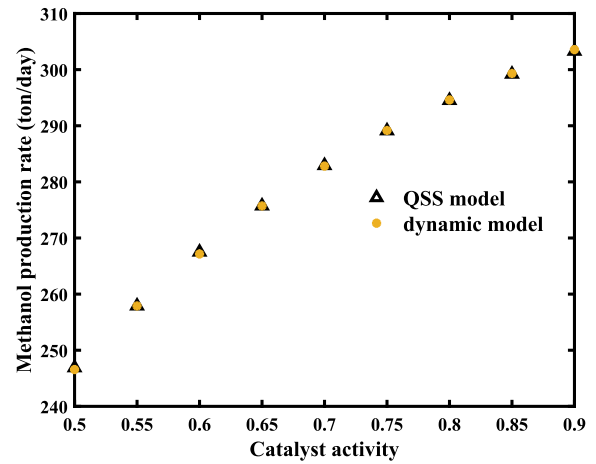


Fig. 5. Comparison of the dynamic and QSS models.

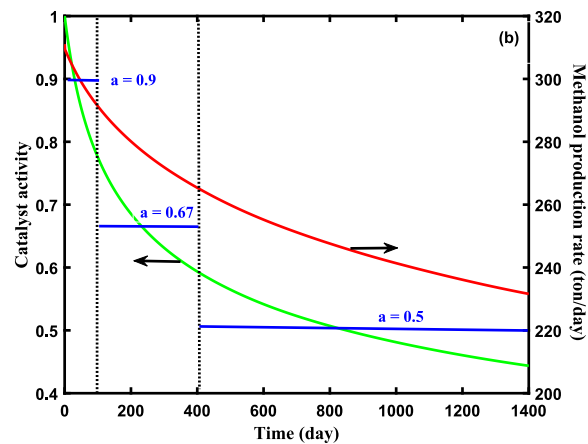
the approach toward thermodynamic equilibrium. The observed temperature profile within the reactor underscores the importance of efficient heat removal through the cooling system for managing the heat load and maintaining optimal operating conditions for methanol synthesis.

Fig. 4(b) illustrates the average catalyst relative activity and methanol production rate over time. Due to the gradual deactivation of the system's catalyst, as shown in the figure, the catalyst's relative activity decreases significantly, leading to a corresponding reduction in methanol production.

To find the best cooling scenario and maintain favorable methanol production conditions over time, despite the catalyst deactivation issue, the catalyst's lifetime is divided into three distinct periods (Fig. 4(b)). The average catalyst activity for each time period is determined by Eq. (21). During the first period (0–100 days, equivalent to an activity factor of $\langle \bar{a} \rangle = 0.9$), the sharp deactivation rate results in a shorter timeframe. The second period (100–400 days, equivalent to $\langle \bar{a} \rangle = 0.67$) sees a gradual decrease in the deactivation slope. Finally, during the third period (400–1400 days, equivalent to $\langle \bar{a} \rangle = 0.5$), the catalyst deactivation occurs at a much milder rate. Beyond 1400 days, the catalyst activity becomes so low that regeneration or replacement is necessary.

$$\langle \bar{a} \rangle = \frac{\int_{t_1}^{t_2} \bar{a} dt}{\int_{t_1}^{t_2} dt} \quad (21)$$

In this study, two models are used to predict the theoretical behavior of the reactor: the simple QSS model and the complex dynamic model.



For the QSS model, the steady-state calculations to determine the methanol production rate are done at certain relative activity values. However, for the dynamic model, catalyst relative activity is calculated at each point along the reactor's length, continuously over time. The QSS model is developed to simplify the dynamic formulation and reduce the computational load. Before using it in the optimization framework, the consistency between the dynamic and QSS models is examined under conventional operating conditions. Fig. 5 compares the methanol production rate predicted by the dynamic model at several time points with that predicted by the QSS model at corresponding catalyst activity levels. As observed, both models provide nearly identical predictions. This good agreement confirms that neglecting the time-derivative terms in the QSS formulation introduces negligible error in the estimated reactor performance, validating its suitability for the subsequent optimization analysis.

Building on the system behavior discussed earlier, the following sections will first examine the operating parameters that influence the optimal bed temperature profile. Subsequently, the most effective strategy for the cooling system design will be proposed.

3.5. Sensitivity analysis

Identifying optimal conditions to maximize product yield remains a central concern for industries. In this regard, the selection of an appropriate cooling system strategy is crucial, as it significantly influences the CO₂ hydrogenation process. Therefore, evaluating the effects of key operating parameters on this process is essential. Following the validation of the proposed model, it can be employed to investigate the system's behavior in terms of the optimal bed temperature profile and methanol mass flow rate. This will facilitate a thorough examination of how various operational parameters impact the hydrogenation process. According to the literature [39,63,64], despite existing structural and operational limitations within the reactor, the most critical parameters affecting the CO₂ hydrogenation process are pressure and the H₂/CO₂ ratio.

As shown in Fig. 6(a), increasing the pressure significantly enhances methanol production within the reactor. This is due to Le Chatelier's principle, which indicates that a rise in pressure shifts the equilibrium toward the side of the reaction with fewer moles of gas, thereby promoting CO₂ consumption and methanol production. At higher pressures, the optimal bed temperature profile shifts toward the end of the reactor to achieve higher reaction rates (see Fig. 6(b)). Nevertheless, increasing the pressure in the range of 50–100 bar does not sharply alter the optimal bed temperature profiles.

Fig. 7 demonstrates that increasing the H₂/CO₂ ratio has a similar effect to raising pressure. As the number of moles in the system

increases, the equilibrium shifts toward the side with fewer moles, favoring greater methanol production. Consequently, increasing the H₂/CO₂ ratio leads to higher methanol yields (see Fig. 7(a)) and a slightly higher optimal bed temperature is required to achieve this improved performance (see Fig. 7(b)).

While increasing pressure and the H₂/CO₂ ratio can substantially improve methanol production, these adjustments introduce several operational challenges. These include the need for a more powerful compressor, elevated costs in the separation unit due to by-product formation, and a higher recycle ratio to manage excess hydrogen. These adjustments also necessitate equipment that can safely handle higher pressures. Consequently, controlling the coolant temperature emerges as the most effective and efficient strategy to increase methanol yield, balancing production gains with operational feasibility.

3.6. Theoretical strategy for obtaining optimal temperature profile

In this section, both two models, QSS and dynamic models, are used to determine the theoretical optimal temperature profiles in the reactor. Based on the dynamic model, for each point in the reactor, the optimal coolant temperature that maximizes methanol production is calculated over given periods, i.e., 0–100, 100–400, and 400–1400 days. This procedure is repeated for every point along the reactor, resulting in the black dashed line graphs shown in Fig. 8.

For the QSS model, the steady-state calculations to determine the optimal temperature profiles for the catalytic bed and cooling side of the reactor are done at certain relative activity values, i.e. 0.9, 0.67, and 0.5. These values represent the average relative activity over a distinct period of the catalyst's lifetime (see Fig. 4).

Fig. 8 shows a comparison between the QSS and dynamic models in terms of optimal catalyst-bed and cooling temperature profiles as well as methanol production rate for both given catalyst relative activities and catalyst lifetime periods. In the first half of the reactor, the highest methanol production rate occurs at the highest allowable temperature of the catalyst bed (i.e., 543 K) due to the speeding up of the forward reaction rate (see Fig. 8(a)). So, at a given catalyst activity or time period, the optimal feed temperature is determined to be 543 K. After that, the catalyst-bed temperature should be reduced to raise the equilibrium conversion. To maintain the maximum allowable temperature of the catalyst bed and due to the reduction of heat produced because of fuel depletion, the optimal temperature of the cooling system inevitably increases (see Fig. 8(b)). After that, the optimal temperature of the cooling system should be gradually reduced to lower the temperature of the catalyst bed.

The sharp temperature decline observed in the optimized profiles marks the transition point where the process shifts from kinetic control

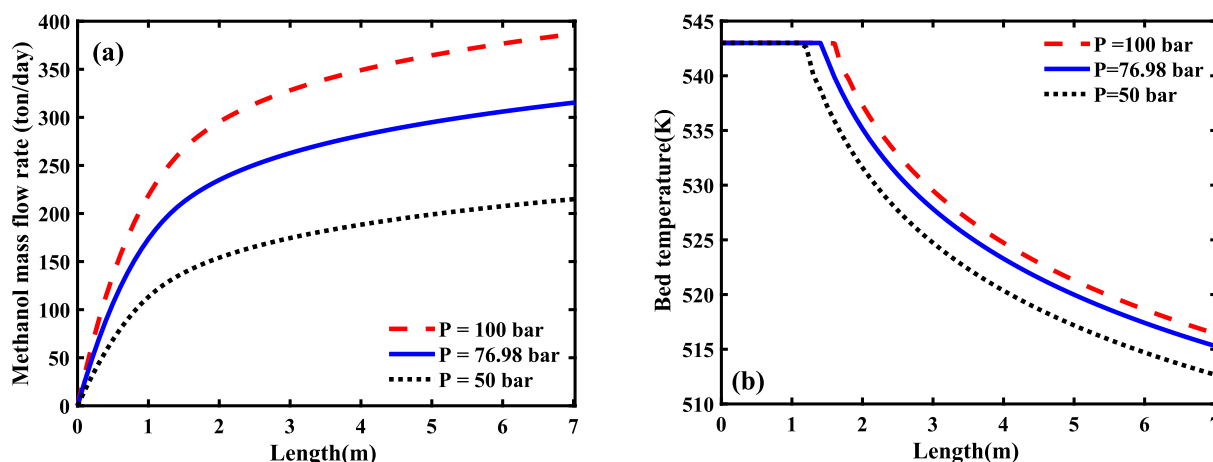


Fig. 6. Effect of pressure on (a) methanol mass flow rate, and (b) optimal bed temperature at steady-state conditions in the presence of fresh catalyst.

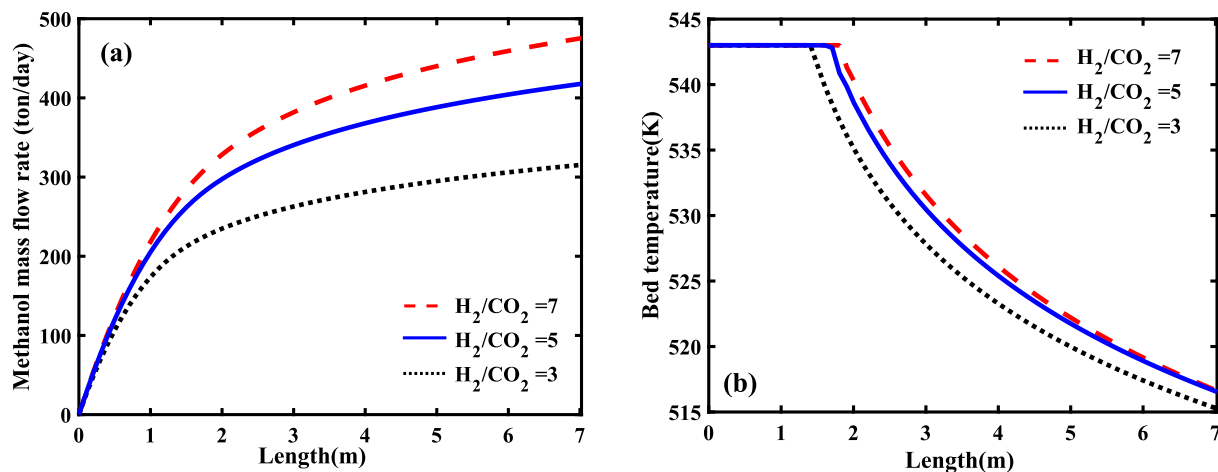


Fig. 7. Effect of molar H_2/CO_2 ratio on (a) methanol mass flow rate, and (b) optimal bed temperature profile at steady-state conditions in the presence of fresh catalyst.

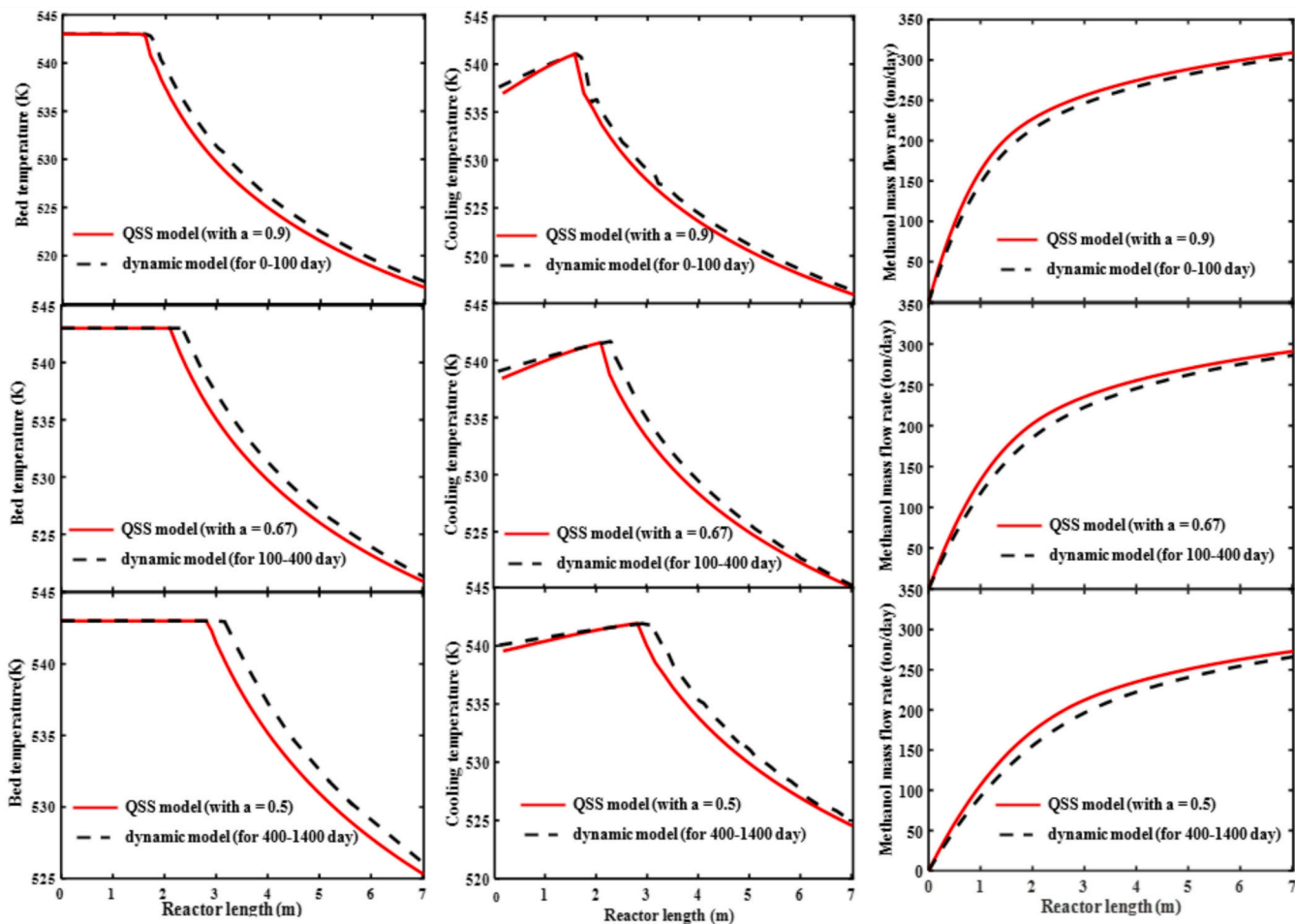


Fig. 8. Optimal (a) bed temperature profile, (b) cooling temperature profile, (c) methanol mass flow rate according to QSS model (with $\tilde{a} = 0.9, 0.67, 0.5$) and dynamic model (for periods of 0–100, 100–400, 400–1400 days).

to thermodynamic control. Upstream of this point, high temperatures maximize reaction rates; downstream, lower temperatures enhance equilibrium conversion. The abruptness of this transition arises from the high temperature sensitivity of both reaction kinetics and thermodynamic equilibrium, creating a well-defined optimal switching point identified by the optimization algorithm.

Compared to Fig. 4(a), the lower bed temperatures observed in the optimized cases (Figs. 6–8) arise from the optimized coolant profile, which reduces the shell-side temperature below 524 K in sections of the reactor to enhance equilibrium conversion.

Fig. 8(a) and (b) indicate that the decrease of the catalyst relative activity from 0.9 to 0.5, or the passing of the catalyst lifetime period

from 0 to 100 to 400–1400 days, shifts both the optimal temperature profiles of the catalyst-bed and cooling system upward and rightward. This issue theoretically arises to offset the reduction in methanol production caused by a decline in catalyst activity (Fig. 8(c)).

Comparing the results of both models shows that the QSS model's ability to predict temperature trends is consistent with the behavior of the dynamic model. Therefore, implementing the QSS model to predict the optimal behavior of the reactor in an industrial plant is more feasible compared to the dynamic approach. This is due to the QSS model's ability to predict a consistent optimal temperature for the cooling system over a fixed period, making it more practical for industrial applications. In contrast, the dynamic model requires continuous adjustments to the coolant temperature at any time, which is technically challenging and economically unviable. In the subsequent Section, more comprehensive details are furnished on this topic.

Table 5 compares the reference case and optimal methanol production rates (in tons per day) for both the QSS and dynamic models across different catalyst activity levels and periods, respectively. The results demonstrate the effectiveness of both models in optimizing methanol production under varying conditions. In the QSS model, as catalyst activity decreases, optimizing the temperature yields progressively higher additional methanol production, with a notable 10.58 % increase at the lowest activity level ($\bar{a} = 0.5$). Similarly, in the dynamic model, the optimization is most impactful in the later periods of the catalyst's lifetime (400–1400 days), resulting in a 7.96 % increase in methanol production. These findings underscore the importance of implementing an optimal cooling temperature strategy, particularly when the catalyst activity is reduced, to significantly enhance methanol production.

3.7. Practical strategy for obtaining optimal temperature profile in a single-stage cooling system

As demonstrated in the previous section, the optimal temperature profile varies not only over time but also along the reactor length, making it impractical to apply these conditions consistently throughout the entire reactor. To overcome this challenge, a practical approach involves implementing a single- or multi-stage cooling system (Fig. 9), where the coolant temperature remains fixed within each stage, ensuring uniformity along the reactor length. For instance, in industrial methanol synthesis reactors, the cooling fluid temperature is typically kept constant (e.g., 524 K in the reference case depicted in Fig. 10(a)) throughout the process duration by using a single-stage cooling system.

An alternative approach to optimizing coolant temperature involves dynamically adjusting it in response to the reactor's real-time conditions (Adaptive model). This dynamic optimization procedure aims to maximize methanol production by identifying the optimal cooling system

Table 5

The additional yield achieved by applying theoretical optimal temperature profile based on QSS and dynamic models.

Model		Methanol production rate (ton/day)		Additional yield (%) [*]
		Reference case	Optimal case	
QSS	Catalyst activity			
	0.9	303.22	309.12	1.94
	0.67	278.65	291.21	4.51
	0.5	246.83	272.94	10.58
Dynamic	Period (day)			
	0–100	300.50	304.50	1.33
	100–400	277.63	286.40	3.16
	400–1400	246.59	266.21	7.96

$$^* \text{ Additional yield} = \frac{\text{Optimal} - \text{Reference case}}{\text{Reference case}} \times 100.$$

temperature at any given time. The theoretical optimal coolant temperature for a single-stage cooling system, represented by the blue line in Fig. 10(a), shows an increasing trend over time as it compensates for the gradual decline in catalyst activity. This adjustment leads to higher methanol production compared to the reference case, as illustrated in Fig. 10(b).

Here, the optimization procedure is applied to a single-stage cooling system using both the QSS and dynamic models. For the QSS model, catalyst activity is fixed at predetermined values of 0.9, 0.67, and 0.5, while the dynamic model considered timeframes of 0–100, 100–400, and 400–1400 days. The optimal coolant temperatures, corresponding to the maximum methanol production rates, are represented by the black and pink lines in Fig. 10. The benefit of adopting such a temperature optimization method is evident in Fig. 10(b), where methanol production is enhanced from 232 to 256 tons per day during the later periods of the catalyst's lifetime. This increase demonstrates the effectiveness of the optimized cooling temperature strategy.

The results show that the cooling temperature profiles from both the dynamic and QSS models are very similar, indicating that the QSS approach provides comparable performance to the dynamic model for the single-stage cooling system. This finding suggests that, unlike more complex theoretical strategies, the simple QSS model can be applied with minimal error compared to the dynamic model. Therefore, the QSS strategy offers a practical and feasible solution for optimizing temperature control in industrial reactors, particularly in multi-stage configurations, without sacrificing production efficiency.

Table 6 summarizes the methanol production (in tons) achieved through QSS and dynamic models within a single-stage cooling system over three distinct periods: 0–100, 100–400, and 400–1400 days. Notably, the QSS model demonstrates significant advantages in the timeframe of 400–1400 days, yielding an 8.76 % additional yield compared to the reference case, with a relative error of only 1.61 % when compared to the dynamic model. This suggests that the QSS model closely approximates the dynamic model's performance while offering a simpler implementation for industrial applications. Furthermore, the results indicate that optimizing the temperature profile, particularly during the later periods of catalyst activity, significantly enhances methanol production efficiency, reinforcing the viability of the QSS model as a practical strategy for optimizing reactor performance.

3.8. Practical strategy for obtaining optimal temperature profile in a multi-stage cooling system

In the quest to enhance methanol production efficiency in multi-stage cooling systems, a practical strategy is proposed to establish an optimal temperature profile throughout the reactor. This strategy leverages the insights gained from both QSS and dynamic models to inform temperature control mechanisms that are responsive to the varying conditions of catalyst activity over time. The approach centers on the implementation of a multi-stage cooling system, which allows for temperature adjustments tailored to distinct operational phases of the catalyst's lifecycle. By employing a systematic optimization procedure, the coolant temperature can be precisely regulated at each stage, ensuring that it aligns with the maximum methanol production potential associated with specific catalyst activity levels. In other words, the optimization process ensures that the coolant temperature remains consistent within each stage while adapting to the changing dynamics of the reaction environment. This results in improved methanol yields, particularly during critical phases of catalyst activity decline, thereby maximizing overall reactor performance.

Fig. 11 illustrates the optimal coolant temperature profiles for each stage of a two-stage cooling system over time. The selected timespans of 0–100, 100–400, and 400–1400 days align with the previous sections, demonstrating a gradual increase in coolant temperatures in both stages. Notably, the first stage requires higher temperatures than the second stage to facilitate the highest methanol production rate within the

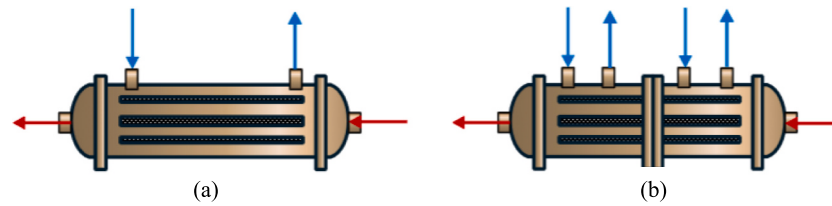


Fig. 9. Typical design of the (a) single-stage, and (b) two-stage cooling system.

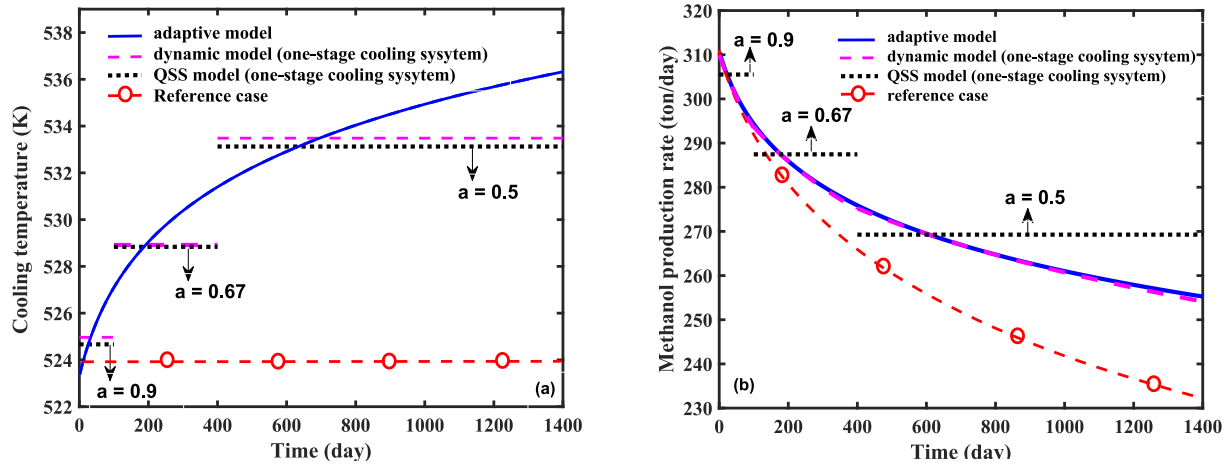


Fig. 10. Comparison of optimal (a) cooling temperature profile, and (b) methanol production rate according to QSS and dynamic models in the single-stage cooling system with those based on reference case.

Table 6

Comparison between QSS and dynamic models in terms of methanol production in a single-stage cooling system.

Period (day)	Methanol production (ton)		Additional yield (%)		Relative error (%) *
	Reference case	Optimal	QSS model	Dynamic model	
0–100	30,049	30,456	1.35	0.34	1.01
100–400	83,287	85,947	3.19	2.26	0.91
400–1400	246,590	268,190	8.76	7.04	1.61

$$* \text{ Relative error} = \frac{\text{QSS model} - \text{Dynamic model}}{\text{Dynamic model}} \times 100$$

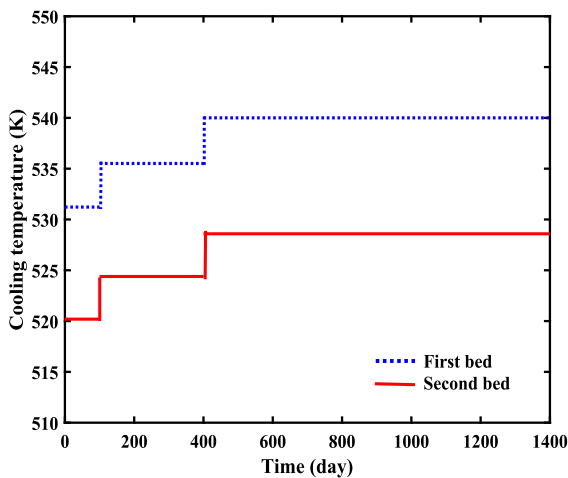


Fig. 11. Optimal coolant temperature in each stage of the two-stage cooling system over time.

Table 7

Optimal coolant temperature and first-bed length of two-stage cooling system.

Catalyst activity	Length of first bed (m)	Coolant temperature (K)		Additional yield (%)
		First bed	Second bed	
0.9	3.21	531.25	520.19	1.71
0.67	3.15	535.73	524.43	4.14
0.5	3.25	539.62	528.63	10.16

catalyst bed. This increase is necessary because higher temperatures accelerate the forward reaction rate, optimizing methanol yield during the initial catalyst activity phase. However, as the catalyst ages, the heat generated decreases, necessitating a controlled rise in the coolant temperature to maintain the catalyst bed's optimal temperature. In contrast, the second stage operates at lower temperatures to enhance equilibrium conversion.

Complementing this analysis, Table 7 presents the optimal coolant temperatures and the corresponding lengths of the first bed in the two-stage cooling system across varying catalyst activity levels. At various

Table 8
Optimization results and process profitability of single- and multi-stage cooling systems.

Catalyst activity	Coolant temperature (K)			Additional yield (%)	Utility cost (M\$)		Gross Revenue (M\$)	
	First bed	Second bed	Third bed		Optimal	Reference case	Optimal	Reference case
Single-stage cooling system: $L = 7.022$ m								
0.9	524.60			0.44	0.22	0.08	11.27	11.22
0.67	528.77			2.81	0.62	0.20	31.81	30.94
0.5	533.09			8.65	1.92	0.55	99.27	91.36
Two-stage cooling system: $L_1 = 3.226$ m, $L_2 = 3.796$ m								
0.9	531.22	520.19		1.71	0.23		11.41	
0.67	535.51	524.39		4.14	0.65		32.22	
0.5	540	528.59		10.16	2.01		100.64	
Three-stage cooling system: $L_1 = 2.5$ m, $L_2 = 2.244$ m, $L_3 = 2.278$ m								
0.9	532.78	524.35	518.21	1.86	0.23		11.43	
0.67	537.29	528.46	522.34	4.38	0.66		32.30	
0.5	540	533.14	526.81	10.35	2.04		100.82	

catalyst activities from 0.9 to 0.5, the optimum length of the first bed remains almost constant, indicating a fixed design for maintaining optimal conditions. However, the coolant temperature for both beds increases with declining catalyst activity, reflecting the need for higher temperatures in the first bed to ensure adequate methanol production. The additional yield percentage highlights the efficiency of the cooling system, with significant increases observed at lower catalyst activity levels, reinforcing the necessity of an adaptable cooling strategy. This combination of temperature optimization and bed length adjustment underscores the importance of a tailored approach in industrial applications, ensuring sustained methanol production throughout the catalyst's operational lifespan.

Table 8 presents the optimization results and process profitability of single- and multi-stage cooling systems across different catalyst activity levels, shedding light on the economic consequences of various cooling strategies in methanol production. As shown in Fig. 4(b), the catalyst activity level remains around 0.5 for the majority of its operational lifespan, making the optimal point at an activity of 0.5 a primary focus. The data demonstrate that as catalyst activity decreases from 0.9 to 0.5, both coolant temperatures and additional yield percentages increase across all configurations, along with corresponding increases in utility costs and revenue.

In the single-stage cooling system, the additional yield increases from 0.44 % to 8.65 % as catalyst activity declines, leading to a similar percentage increase in gross revenue compared to reference case. At the lowest catalyst activity level, the significant increase in gross revenue from \$91.36 million to \$99.27 million (i.e., 8.66 %) far exceeds the increase in utility costs from \$0.55 million to \$1.92 million, indicating that this optimal cooling strategy is highly profitable even within a simplified cooling system.

Based on the lengths derived from the optimization at various activity levels, and considering the respective durations of the process's operating times, the optimal lengths for the first and second stages are adjusted using time-period-based weight factors. Following this adjustment, the optimal lengths for the first and second stages are 3.226 m and 3.796 m, respectively (see Table 8).

In the two-stage cooling system, utilizing variable coolant temperatures for each bed further improves yield and energy management. At a catalyst activity of 0.5, the additional yield reaches 10.16 %, with utility costs totaling \$2.01 million, resulting in gross revenue of \$100.64 million. Compared to the reference case, despite the rise in utility costs from \$0.55 million to \$2.01 million, the substantial increase in gross revenue from \$91.36 million to \$100.64 million (i.e., 10.16 %) demonstrates the economic value of a two-stage cooling approach to maximize both yield and profitability.

The three-stage cooling system, with optimal lengths of $L_1 = 2.5$ m, $L_2 = 2.244$ m, and $L_3 = 2.278$ m, achieves the highest additional yield of 10.35 % at the lowest catalyst activity. Here, utility costs increase to

\$2.04 million, with a marginal revenue increase of \$100.82 million. Compared to the reference case, this represents a 10.35 % increase in gross revenue (from \$91.36 million to \$100.82 million).

Initial estimates indicate that, compared to the reference case, the economic potential (defined as the difference between total gross revenue and total utility costs) increases by \$6.9 million for the single-stage cooling system, \$8.69 million for the two-stage system, and \$8.93 million for the three-stage system over a 4-year operational period. These results suggest that the additional economic potential achieved by the three-stage system relative to the two-stage system is marginal. On the other hand, the capital costs of the three-stage system are expected to be significantly higher than the two-stage system. Consequently, from an economic perspective in this conceptual design phase, the two-stage cooling system remains the more cost-effective option due to its favorable balance between additional yield and estimated operating costs.

It is essential to note that while utility costs increase due to higher energy requirements, the substantial rise in additional yield effectively compensates for these expenses across all configurations. This illustrates the economic viability of implementing advanced cooling strategies in industrial methanol production from CO₂ hydrogenation, emphasizing that the enhanced yield and revenue justify the increased utility costs associated with maintaining optimal temperature profiles. Consequently, the adoption of these advanced cooling strategies is not only technically feasible but also economically advantageous, ensuring improved production efficiency and profitability.

3.9. Pricing sensitivity analysis

To assess the robustness of the profitability of the proposed cooling configurations under market fluctuations, a sensitivity analysis was performed by varying the prices of methanol, water, natural gas, and electricity as the key economic parameters by ± 20 % from their base-case values, based on the uncertainties reported in [65,66]. Fig. 12 illustrates the resulting deviations in economic potential relative to the reference case across all economic scenarios for single-, two-, and three-stage cooling systems.

As shown in Fig. 12(a), variations in methanol price exert the strongest influence on the economic potential in all configurations. A 20 % increase in methanol price substantially enhances profitability, while a 20 % decrease causes the sharpest decline.

Among the utilities, variations in natural gas price exhibit the most pronounced effect on economic potential (4–5 %) for all configurations, reflecting its important role as an energy source (Fig. 12(c)). In contrast, changes in water and electricity prices lead to negligible changes in economic potential, less than 0.3 % (Fig. 12(b)) and 0.1 % (Fig. 12(d)), respectively. This is because the costs associated with water and electricity consumption constitute only a small fraction of total operating costs.

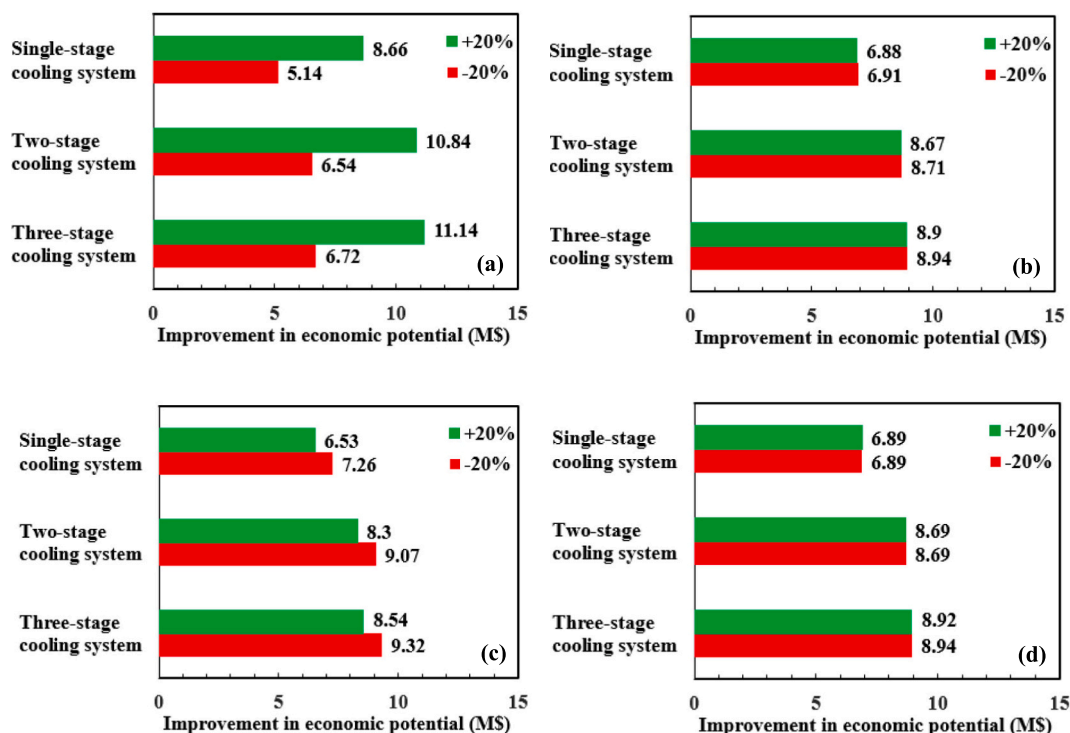


Fig. 12. Effect of $\pm 20\%$ changes in (a) methanol price, (b) water price, (c) natural gas price, and (d) electricity price on improving economic potential

Overall, the sensitivity analysis shows that the economic results are robust over a wide range of market conditions and, consistent with the earlier results, the two-stage cooling system remains the most economically favorable configuration under all tested conditions, even under unfavorable market scenarios (-20% methanol price and $+20\%$ utility costs).

It should be noted that the present economic analysis does not include the capital expenditure associated with multi stage systems—such as costs for additional equipment, sensors, controls, and structural modifications—which represents a limitation of the current study.

4. Conclusion

This study underscored the significance of optimizing the hydrogenation of CO_2 to methanol by applying an optimal temperature profile within a multi-stage cooling system. In this regard, two comprehensive approaches, QSS and dynamic models, were considered. By employing a one-dimensional pseudo-homogeneous mathematical model in the presence of catalyst deactivation, the research accurately represented the concentration and temperature profiles within the catalytic bed, facilitating a detailed understanding of the system dynamics. The following key findings are drawn from this study:

- The QSS model, providing a practical and cost-effective approach, delivered similar outcomes to the dynamic model, facilitating consistent temperature control, particularly for multi-stage cooling systems.
- At a catalyst activity of 0.5, applying the optimal temperature profile throughout the reactor resulted in a 10.58 % increase in methanol production rate compared to conventional reactor. This improvement can also be practically achieved using multi-stage cooling systems.
- Although, over a 4-year operational period, the three-stage cooling system showed a marginal economic potential increase of \$0.24 million compared to the two-stage cooling system, its expected capital costs are significantly higher, making the two-stage cooling

system the more cost-effective choice from an economic perspective at this conceptual design phase.

- Economic analysis reveals substantial revenue enhancements due to applying the optimal two-stage cooling strategy—up to \$144.3 million over 4 years— compared to the conventional process (i.e., \$133.5 million)
- While utility costs increase due to higher energy requirements, the substantial rise in additional yield effectively compensates for these expenses across all configurations of the reactor. This illustrated the economic viability of implementing advanced cooling strategies in industrial methanol production from CO_2 hydrogenation.

Overall, this work provides valuable insights into effective temperature management strategies that maximize methanol production efficiency while accommodating the dynamics of catalyst performance. The findings contribute to the broader field of sustainable energy solutions by advancing operational frameworks for industrial methanol production.

Future research could focus on a comprehensive techno-economic assessment, including the additional capital cost associated with the implementation of multi-stage cooling systems, as well as the potential challenges related to process control and system integration. Such an assessment would help determine whether the technical advantages observed at the conceptual design level can be translated into economically viable solutions at an industrial scale.

CRedit authorship contribution statement

Maedeh Afzali: Validation, Software, Methodology, Investigation. **Mohammad Hasan Khademi:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Investigation, Conceptualization. **Mahboubeh Toghyani:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Appendix

To demonstrate that the algorithm reaches convergence before the termination criterion (150 iterations) is met, the evolution of the objective function value versus the number of iterations is presented in Fig. A.1. The convergence plots are provided for two representative cases at the first node: Fig. A1(a) corresponds to the QSS model (with $\tilde{\alpha} = 0.9$), and Fig. A1(b) to the dynamic model (for 0–100 days). The stabilization of the objective function value after a certain number of iterations confirms that the obtained solutions represent true optima and are not artifacts of premature termination.

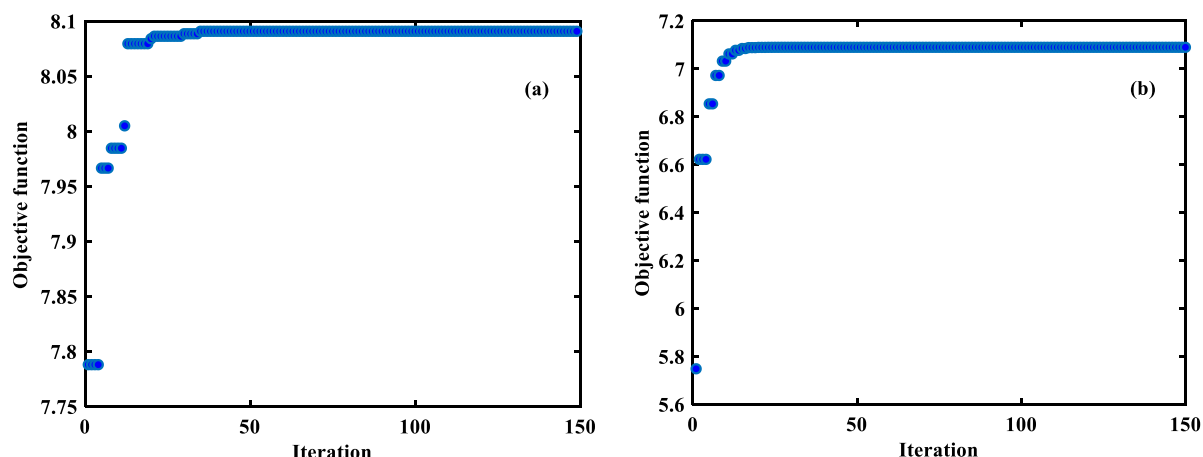


Fig. A.1. Variations of objective function versus number of iterations for (a) the QSS model (with $\tilde{\alpha} = 0.9$), and (b) the dynamic model (for 0–100 days) at the first node.

Data availability

Data will be made available on request.

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